Component ablation — read the vector, never one number

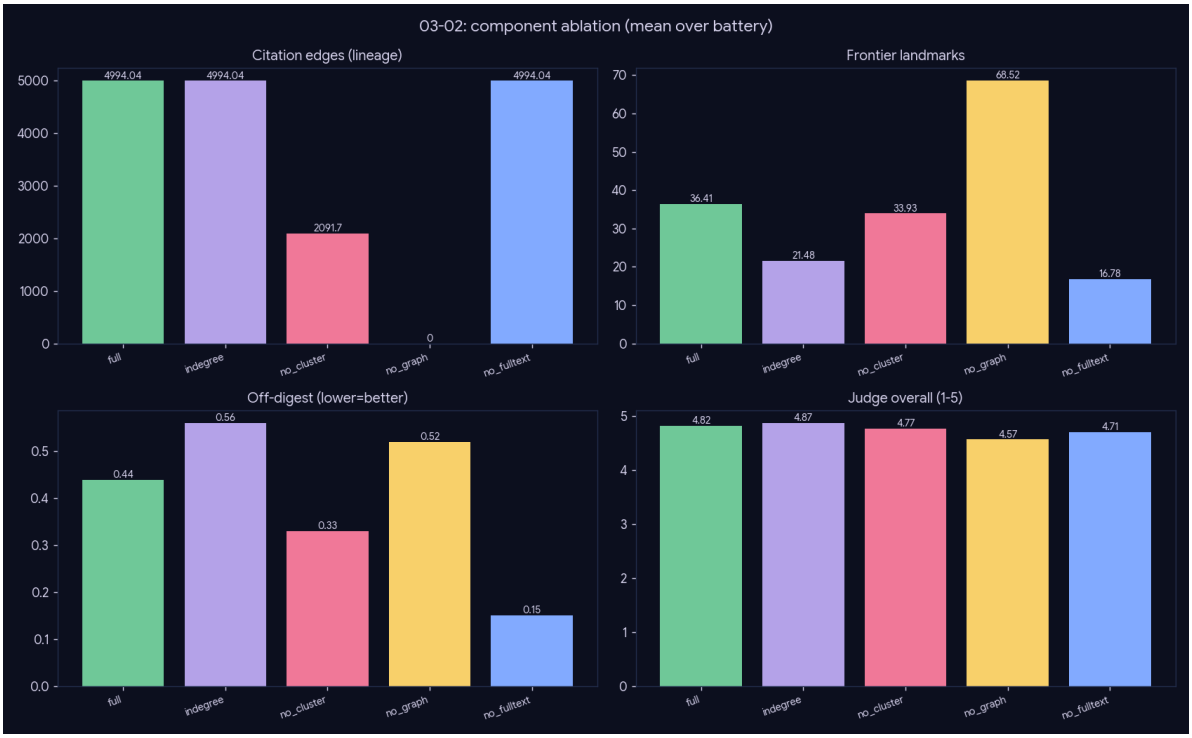
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What we did

The classic ablation: knock out one pipeline component at a time (control = `full`) and read the *full metric vector* + an independent 4-axis LLM judge (Gemma-4-26B: grounding / coverage / lineage / coherence) + a faithfulness tripwire, over the 27-topic battery. Conditions: `indegree` (raw in-degree, no PPR/frontier reservation), `no_cluster` (drop the topic-tree leaf anchor), `no_graph` (vector-only, no citation subgraph), `no_fulltext` (abstracts only). The point is to reproduce the legacy finding that **no single metric can be trusted**.

Results (mean over 27 topics)

config	n landmarks	n edges	frontier count	grounded	off digest	num density	landmark recall	j
full	80.0	4994.04	36.41	26.74	0.44	49.82	0.09	
indegree	80.0	4994.04	21.48	26.74	0.56	49.98	0.1	
no_cluster	79.33	2091.7	33.93	26.67	0.33	48.81	0.09	
no_graph	80.0	0.0	68.52	24.59	0.52	45.51	0.01	
no_fulltext	40.0	4994.04	16.78	23.7	0.15	47.81	0.09	



Conclusion

`no_graph` **is the gameable-single-metric artifact**: it posts the *highest* frontier count (68.5, pure similarity floods recent papers) yet **0 citation edges**, near-zero landmark recall, and the *worst* judge overall (4.57) — chronology and canonical coverage are destroyed while a surface metric improves. `indegree` starves the frontier (21.5 vs 68.5 structure; full = healthy middle) — raw in-degree favors old hubs, the age-vs-citation bias PPR+frontier reservation fixes. `no_cluster` cuts citation density (4994 $\rightarrow$ 2092 edges): the topic-tree leaf anchor keeps the pool connected and on-topic. `no_fulltext` keeps structure but halves landmarks and lowers the judge — abstracts carry less to synthesize. **Faithfulness** ≈ 0.95 **everywhere** — a correctness tripwire, not a lever-ranking score. **The thesis:** the full system is the only configuration that keeps *all* axes healthy at once; any knockout that wins one metric loses others catastrophically. **Caveat:** judge-overall is compressed ($\sim$4.6–4.9, the known distribution collapse) and landmark_recall is diluted by the 24 non-gold topics — the robust signals are edges, frontier balance, and the cross-axis pattern.